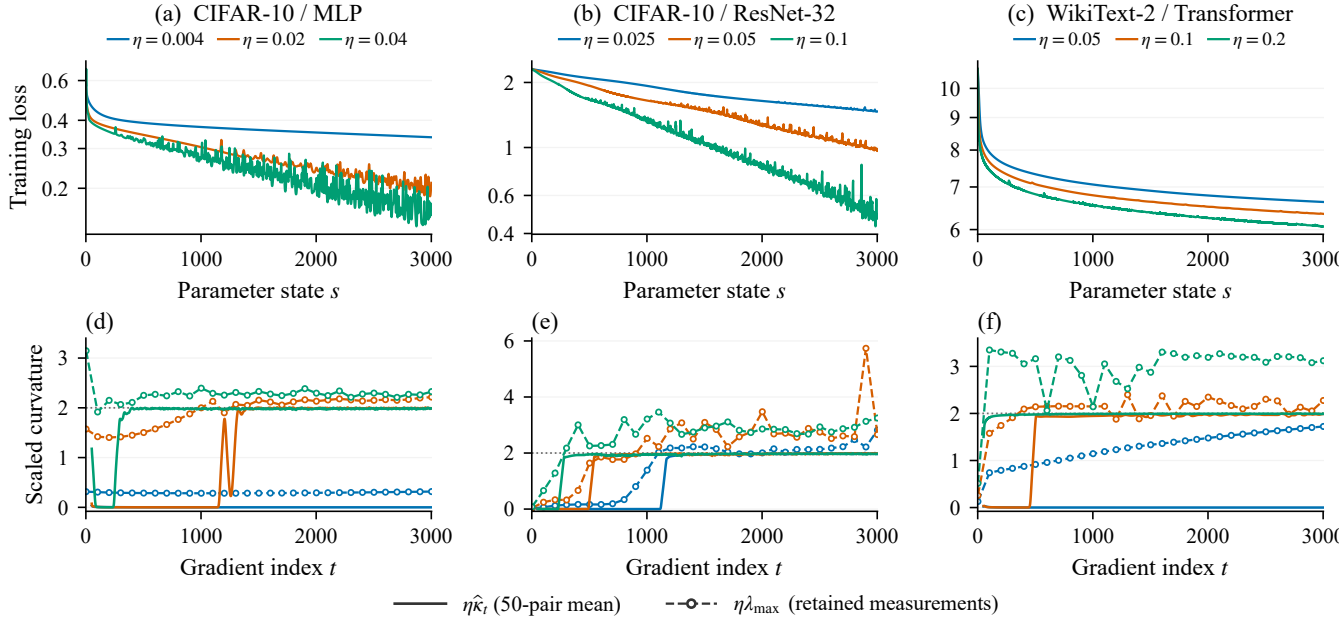

(a) CIFAR-10 / MLP
$\eta = 0.004$ $\eta = 0.02$ $\eta = 0.04$
Training loss
0.6
0.4
0.3
0.2
0
1000
2000
3000
Parameter state s
(b) CIFAR-10 / ResNet-32
$\eta = 0.025$ $\eta = 0.05$ $\eta = 0.1$
2
1
0.6
0.4
0
1000
2000
3000
Parameter state s
(c) WikiText-2 / Transformer
$\eta = 0.05$ $\eta = 0.1$ $\eta = 0.2$
10
9
8
7
6
0
1000
2000
3000
Parameter state s
(d)
Scaled curvature
3
2
1
0
0
1000
2000
3000
Gradient index t
(e)
6
4
2
0
0
1000
2000
3000
Gradient index t
(f)
3
2
1
0
0
1000
2000
3000
Gradient index t
$\eta\hat{\kappa}_t$ (50-pair mean)
$\eta\lambda_{\max}$ (retained measurements)